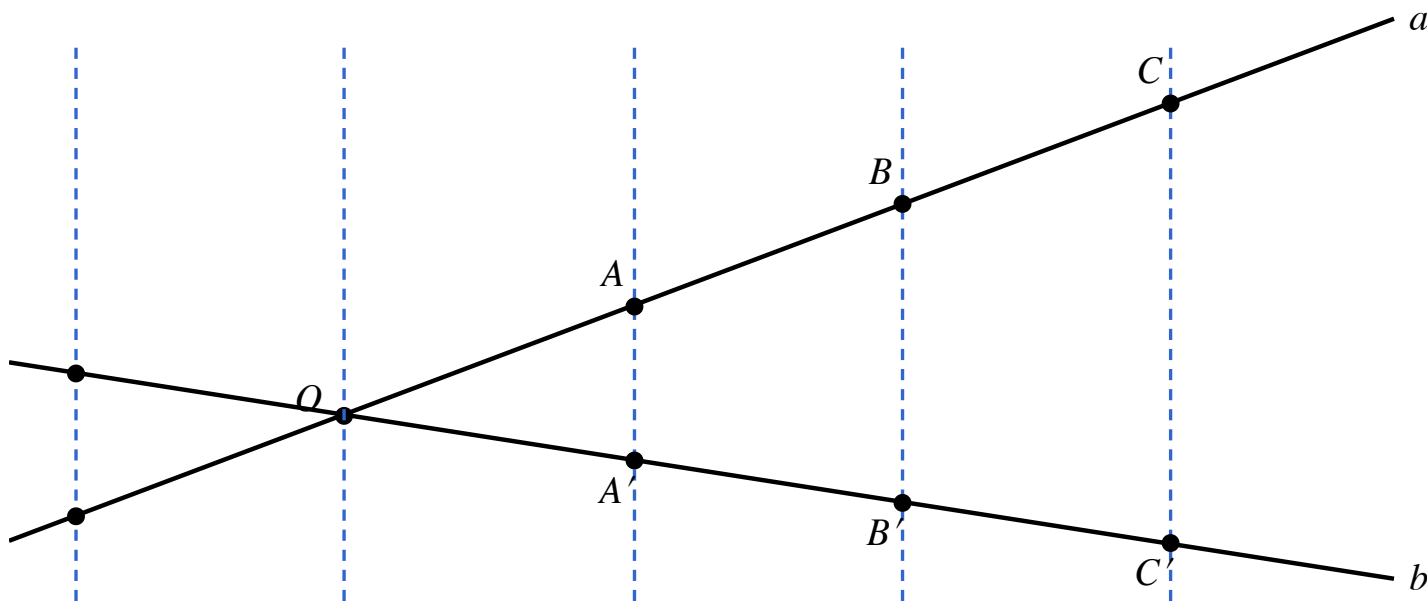


Lecture 6: Trapezoids and Medians

Parallel projection

Consider two lines a and b intersecting at point O . Any family of parallel lines can be used to project points from line a to line b . We call this process a **parallel projection**.

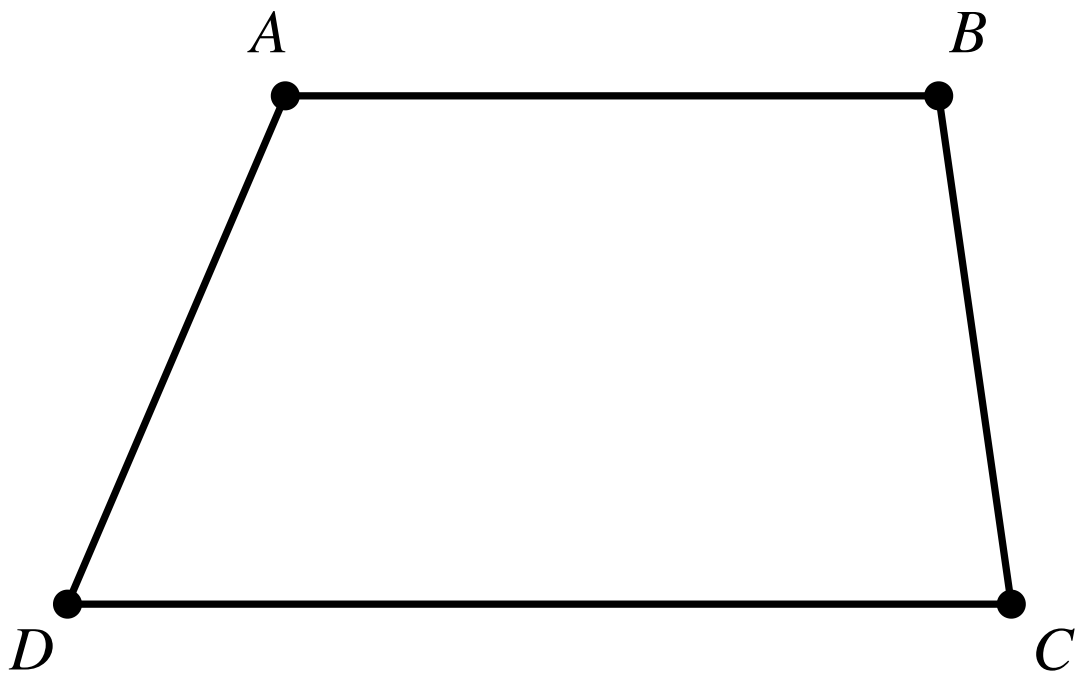


Here is the restated important lemma about parallel projection, where we proved in last lecture using properties of parallelograms.

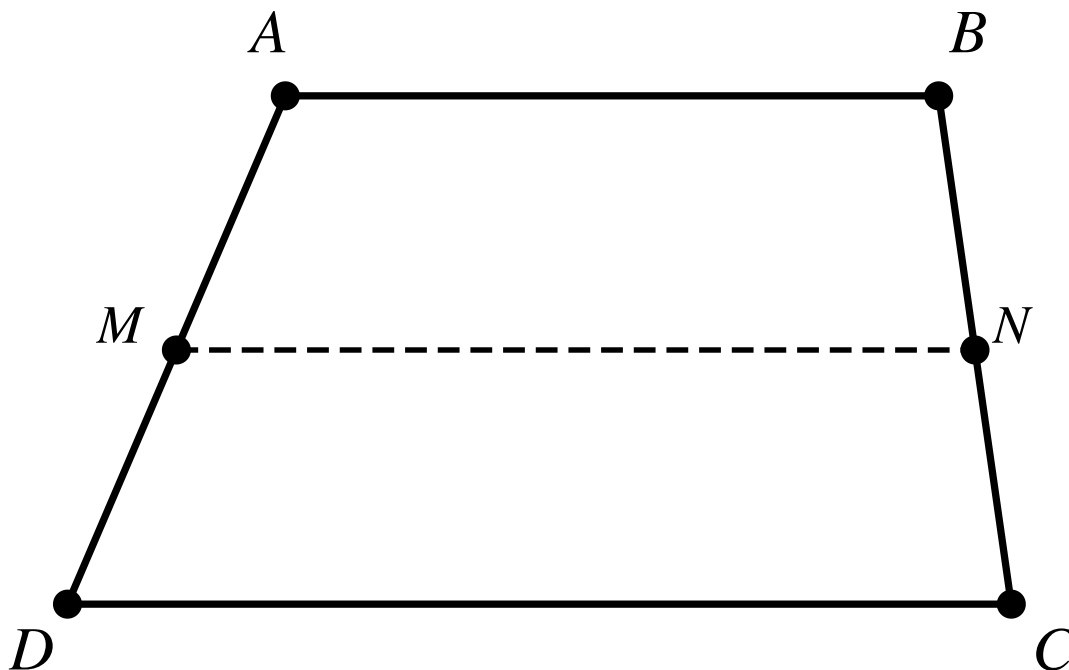
Key Lemma: Equal segments gives equal segments under parallel projection.

Trapezoids

A **trapezoid** is a quadrilateral with at least one pair of parallel sides.



The **median** of a trapezoid is the line segment that passes through the mid point of AD and parallel to AB or CD .



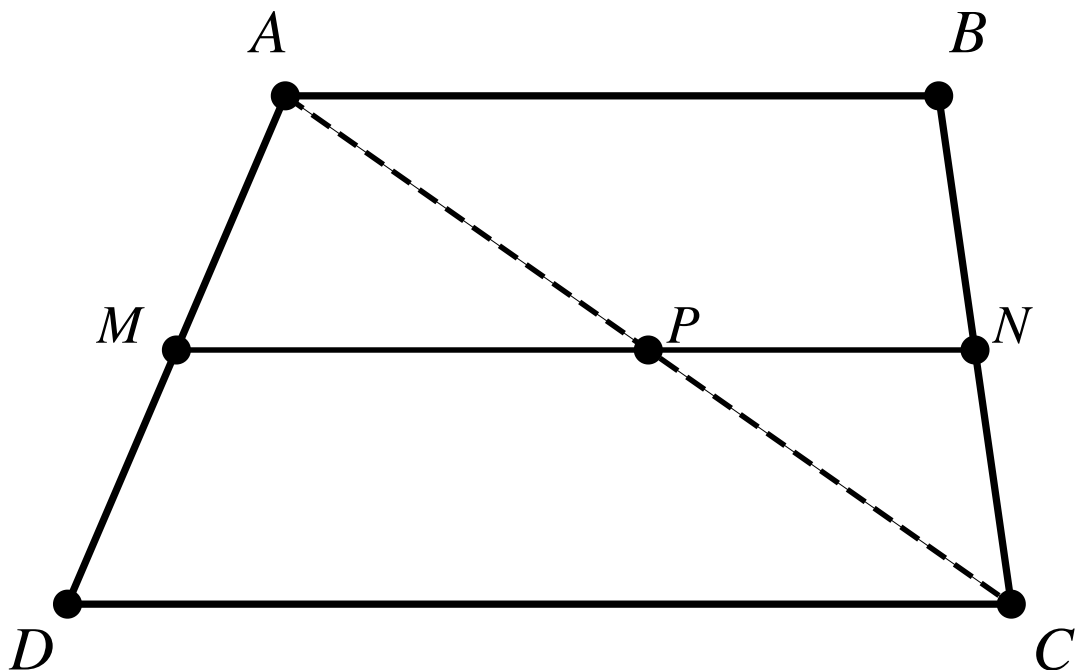
By the key lemma of parallel projection, the median intersects BC at its mid point. So the median is equivalently defined as the line segment that connects the mid points of AD and BC .

Theorem the length of the median is equal to the average of the lengths of the two parallel sides, i.e.,
 $\text{median} = \frac{AB+CD}{2}$.

Proof:

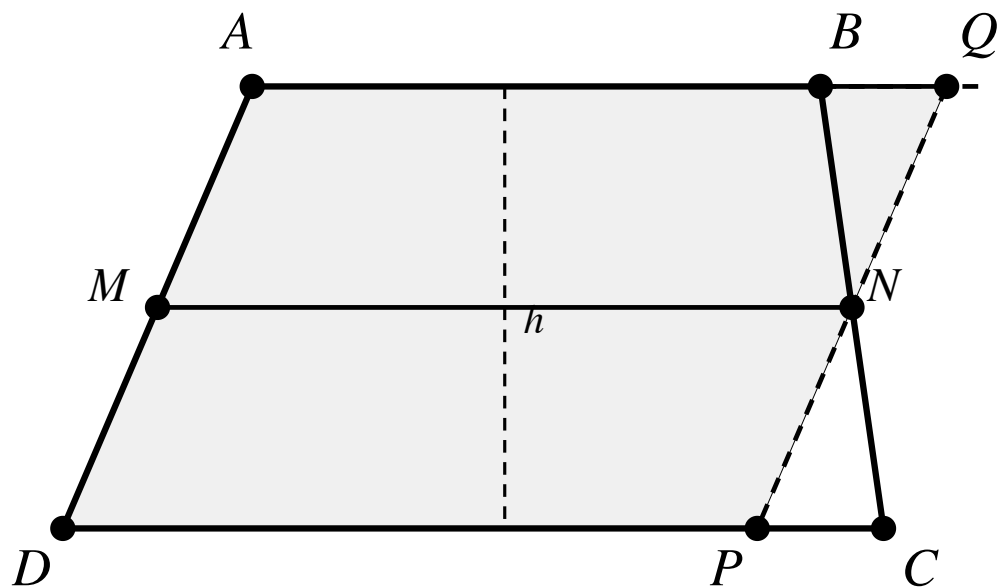
1. We connect A and C and it intersects MN at P .
2. By the key lemma of parallel projection, we know that $AP = PC$. In particular, MP is the bimedial of triangle ADC and PN is the bimedial of triangle ACB .
3. By the bimedial theorem, we know that $MP = \frac{AD}{2}$ and $PN = \frac{BC}{2}$.
4. So

$$MN = MP + PN = \frac{AD}{2} + \frac{BC}{2} = \frac{AD + BC}{2}$$



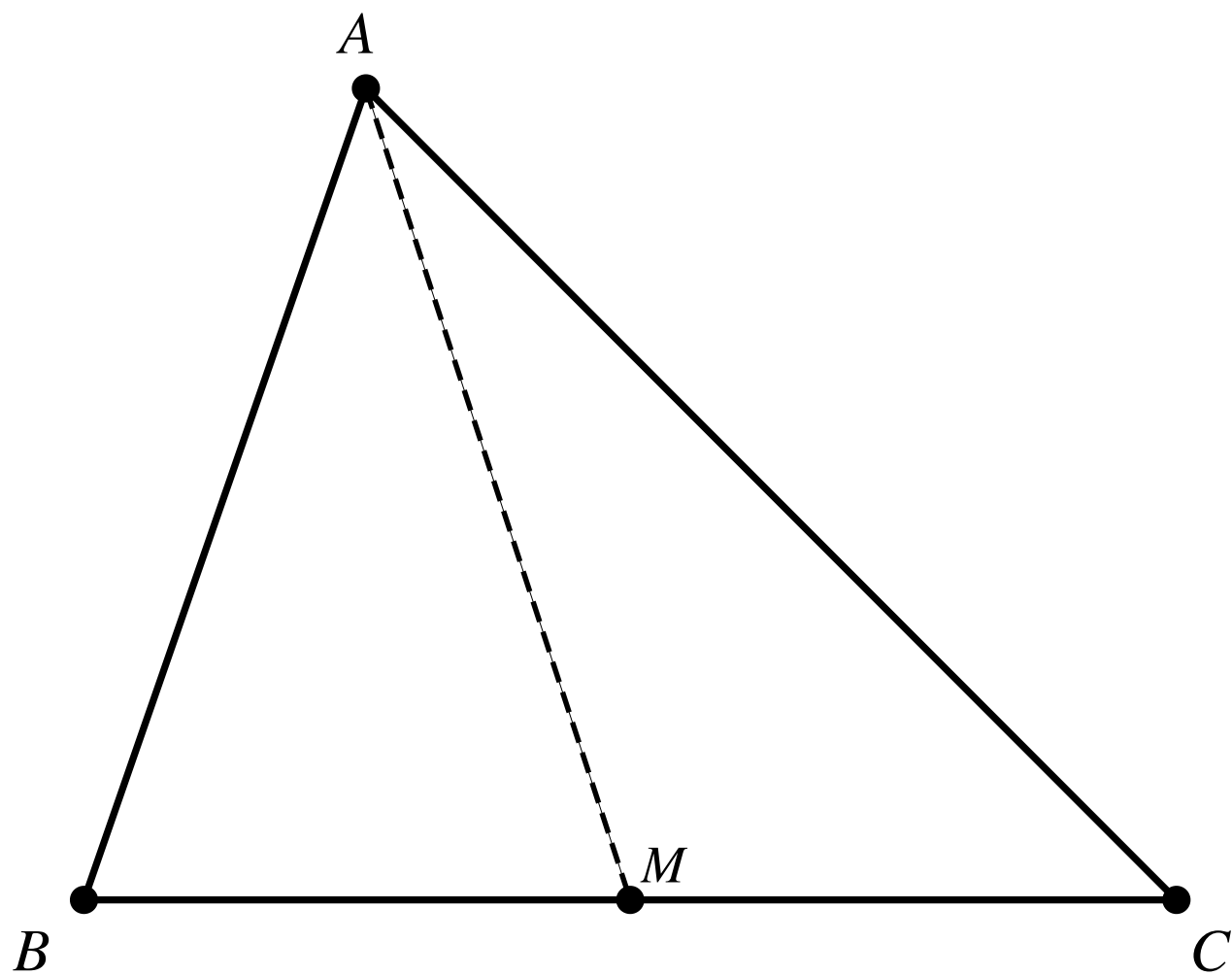
The above formula actually related to the area of the trapezoid as well. We can construct a line through N parallel to AD and let it intersect the extension of AB at Q and DC at P . Then $ADPQ$ is a parallelogram and its area is the same as the area of trapezoid $ABCD$. So the area of trapezoid $ABCD$ is given by

$$\text{Area} = \text{median} \times \text{height} = \frac{AB + CD}{2} \times \text{height}.$$

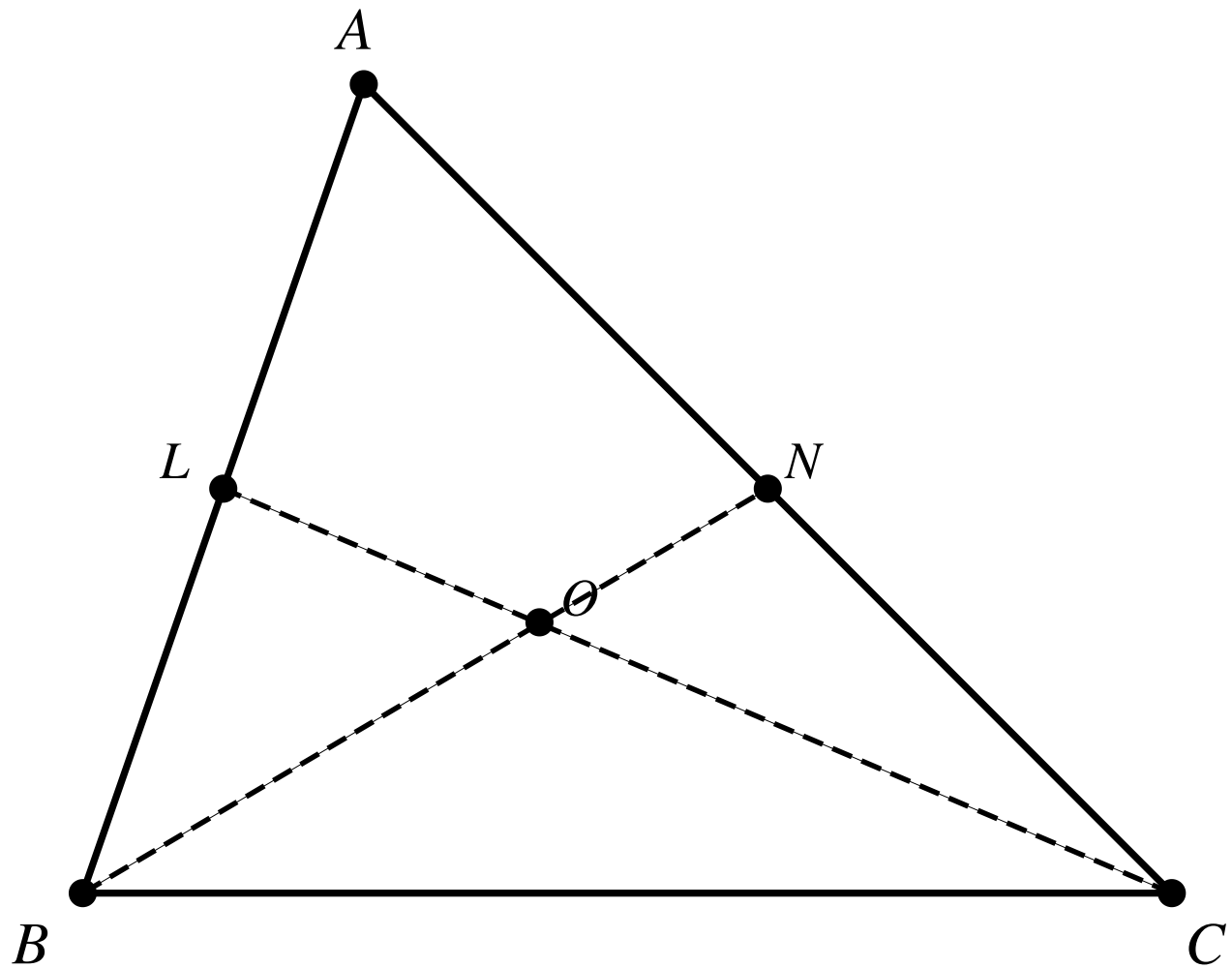


Medians of a triangle

A **median** of a triangle is the line segment that connects a vertex to the mid point of the opposite side.

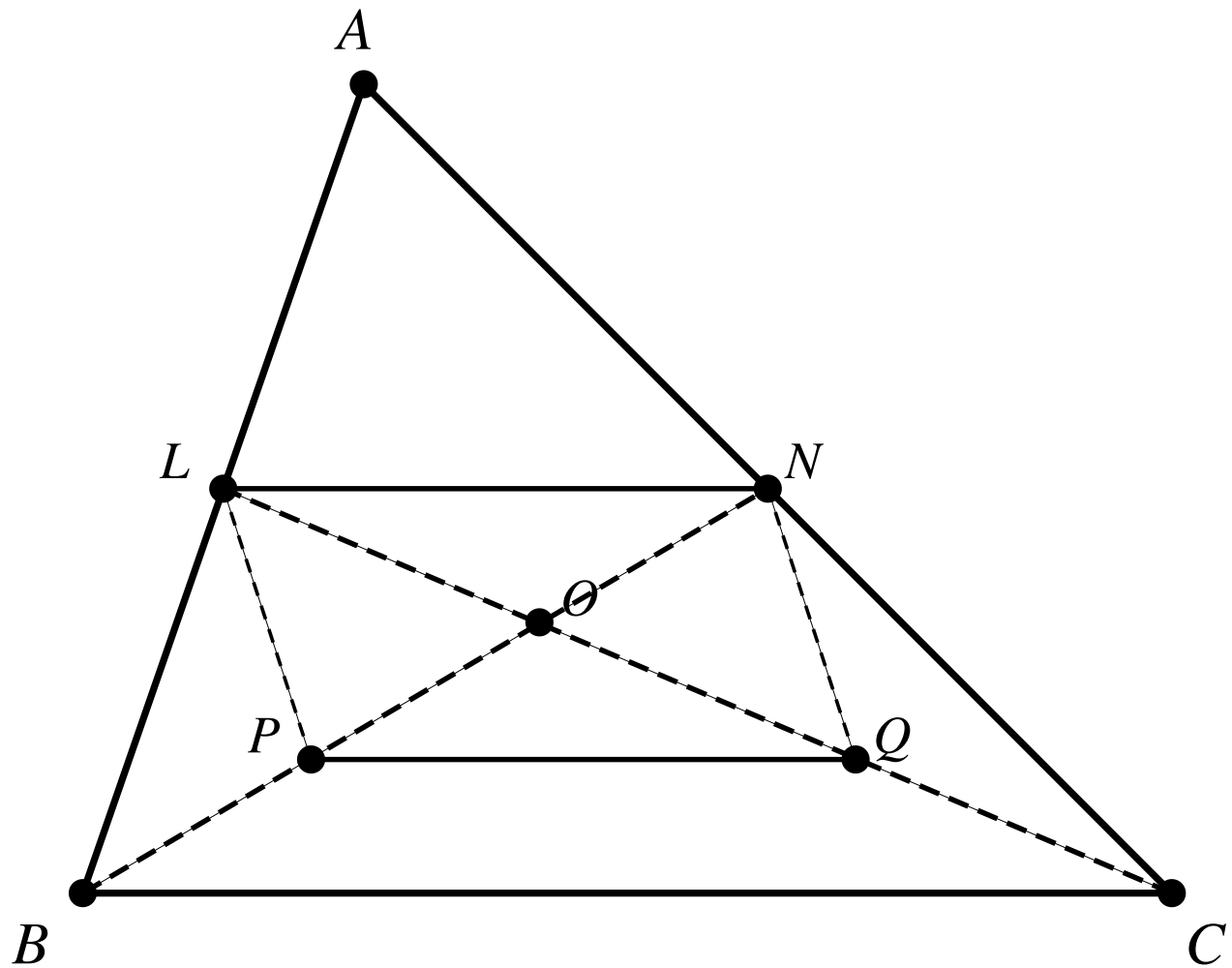


Theorem: The two medians BN and CL intersect at point O and $BO : ON = LO : OC = 2 : 1$.



Proof:

1. Construct the bimedial LN of triangle ABC . So we have $LN = \frac{1}{2}BC$ and $LN \parallel BC$.
2. Construct the bimedial PQ of the triangle BCO which is parallel to BC . We know that $PQ = \frac{1}{2}BC$ and $PQ \parallel BC$.
3. By transitivity, we have $LN = PQ$ and $LN \parallel PQ$. That tells us $LNPQ$ is a parallelogram.
4. Since NP and LQ are the diagonals of the parallelogram $LNPQ$, they bisect each other. So O is the mid point of both NP and LQ .
5. Thus we have $NO = OP = PB$ and $LO = OQ = QC$. The theorem is proved.



Theorem clearly works for any two medians of the triangle. So AM and CL also intersect at point O and $AO : OM = LO : OC = 2 : 1$. This tells us that all three medians of the triangle intersect at the same point O , which is called the **centroid** of the triangle.

Corollary: The three medians of a triangle intersect at the centroid which is two thirds the distance from each vertex to the mid point of the opposite side.

